

DEXCOPHAN COUGH LINCTUS PLUS

VIDEX07-VAR (MY)

DESCRIPTION

Dark brown liquid with blackcurrant flavour.

COMPOSITION

Each 5 ml contains:	
Dextromethorphan HBr	10.00 mg
Ephedrine HCl	6.25 mg
Ammonium Chloride	150.00 mg

PHARMACODYNAMICS

Antitussive: Dextromethorphan has a depressant action on the cough centre similar to that produced by codeine. It acts centrally to elevate the threshold for coughing.

Bronchodilator: Ephedrine relaxes bronchial smooth muscle by stimulation of beta-2-adrenergic receptors, thereby relieving mild bronchospasm, increasing vital capacity, improving air-exchange, and decreasing residual volume. It may also inhibit antigen induced release of histamine.

Vasoconstrictor: Ephedrine acts on alpha-adrenergic receptors of blood vessels in the nasal mucosa, producing vasoconstriction which may result in nasal decongestion.

NH4Cl Ammonium Chloride Expectorant: The ammonium ion supposedly exerts an expectorant action. Its action is doubtful as to whether its irritant action on the gastric mucous membrane contributes to any expectorant action.

Pharmacokinetics

Absorption: Dextromethorphan is fully absorbed from the gastrointestinal tract. Ephedrine is well absorbed when taken orally. Ammonium chloride is rapidly absorbed from the gastro-intestinal tract.

Blood Concentration: After an oral dose of 30mg of dextromethorphan HBr, 10%, 25% and 10% of the dose was detected in the blood at 1, 2.5 and 6 hours respectively after administration. Nevertheless plasma levels of unmetabolised dextromethorphan following the administration of therapeutic oral doses are very low. This is due to extensive metabolism in the liver (first pass effect) and distribution which remove the drug rapidly from the blood.

When 22mg of ephedrine HCl oral dose is given, a peak plasma concentration of 40 to 140ng/ml are attained.

Half-Life: Plasma half-life for ephedrine HCl is 3 to 4 hours.

Distribution: Ephedrine is accumulated in the liver, lungs, kidneys, spleen and brain.

Metabolic Reactions: After oral administration, dextromethorphan is absorbed from the intestinal tract and passes via the portal vein to the liver before entering the general circulation. In the liver, it is metabolised (first pass effect) to three metabolite. There is a wide variation between the individuals in the large first pass effect. Therefore, the proportions of dextromethorphan that are O-demethylated an N-demethylated vary widely between individuals. After O- or N-demethylation, the metabolites undergo glucuronic acid or sulphate conjugation.

Ephedrine undergoes N-Demethylation and oxidative deamination followed by conjugation.

The ammonium ion is converted into urea in the liver; the anion thus liberated into the blood stream and extra cellular fluids causes a metabolic acidosis and decreases the pH of the urine.

Excretion: Up to 56% of a dextromethorphan dose is excreted in the urine, about 8% being excreted unchanged in 6 hours. However, there is a wide variation in the reported excretion of dextromethorphan and its metabolites. This reflects the large intersubject differences in the first pass effects.

About 95% of ephedrine may be excreted in the urine in 24 hours.

The rate of urinary excretion of ephedrine is pH- dependent and is increased in acid urine.

Ammonium chloride, as its urea metabolite is excreted through the urine.

INDICATIONS

For the control of cough and alleviation of nasal congestion.

CONTRAINDICATIONS

Not recommended in hypersensitive patients and concurrent monoamine oxidase (MAO) inhibitor therapy.

WARNINGS AND PRECAUTIONS

- Caution in patients with asthma, hepatic or renal function impairment, cardiovascular disease, diabetes mellitus, glaucoma, hypertension, hyperthyroidism, prostatic hypertrophy.
- Not to be used in children less than 2 years of age.
- To be used with caution and doctor's / pharmacist's advice in children 2 to 6 years of age.

PREGNANCY AND LACTATION

Safety for use in pregnancy and lactation has not been established.

MAIN SIDE/ADVERSE EFFECTS

- Nausea or vomiting, dizziness, headache, loss of appetite, gastrointestinal disturbances.
- The common adverse effects of ephedrine are tachycardia, anxiety, restlessness and insomnia. Tremor, dry mouth, impaired circulation to the extremities, hypertension and cardiac arrhythmias may also occur.

OVERDOSAGE

Clinical features: Nausea, vomiting, abdominal discomfort, dizziness, hallucinations, tachycardia, respiratory depression.

Treat overdosage by emesis or gastric lavage, if appropriate, and the necessary symptomatic and supportive measures.

DRUG INTERACTIONS

Concurrent use with the following drugs requires careful monitoring:

General anaesthetics; other sympathomimetics; ergonovine, methylegonovine, oxytocin; guanethidine.

DOSAGE AND ADMINISTRATION

Adults:

Oral, 10ml every four hours as needed, not exceeding 60ml/24 hours.

Children:

2 to 6 years - Oral, 2.5ml every four hours as needed, not exceeding 15ml/24 hours.

7 to 12 years - Oral, 5ml every four hours as needed, not exceeding 30ml/24 hours.

Note: The information given here is limited. For further information, consult your doctor or pharmacist.

Storage: Store below 30°C. Protect from light.

Presentation/Packing: Bottle of 60ml, 100 ml and 120 ml.

Product Registration Holder / Manufactured by: **HOVID Bhd.**
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30010 Ipoh, Perak, Malaysia.

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